Deployment & Instance Management

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AWS Elastic Beanstalk Overview



- Elastic Beanstalk is a developer centric view of deploying an application on AWS

- It uses all the component's we've seen before:
  EC2, Auto Scaling Group, Elastic Load Balancers, RDS, etc…
- But it's all in one view that's easy to make sense of!
- **We still have full control over the configuration of each component**

- Beanstalk is free but you pay for the underlying instances

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Elastic Beanstalk

- Support for many platforms:
  - Go
  - Java SE
  - **Java with Tomcat**
  - .NET on Windows Server with IIS
  - Node.js
  - PHP
  - Python
  - Ruby
  - Packer Builder
  - **Single Container Docker**
  - **Multicontainer Docker**
  - **Preconfigured Docker**
- Beanstalk is great to "**Replatform**" your application from on-premises to the cloud

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Elastic Beanstalk

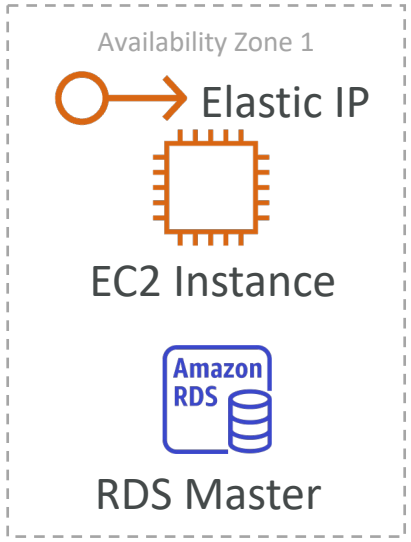
- Managed service
  - Instance configuration / OS is handled by Beanstalk
  - Deployment strategy is configurable but performed by Elastic Beanstalk
- Just the application code is the responsibility of the developer
- Three architecture models:
  - Single Instance deployment: good for dev
  - LB + ASG: great for production or pre-production web applications
  - ASG only: great for non-web apps in production (workers, etc..)

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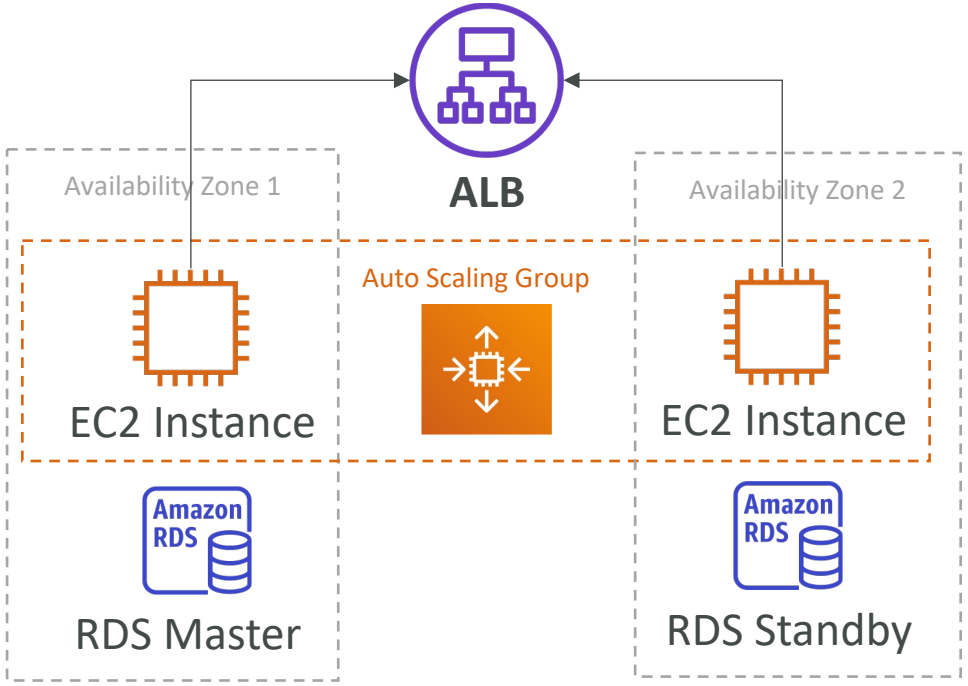
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Beanstalk Environments

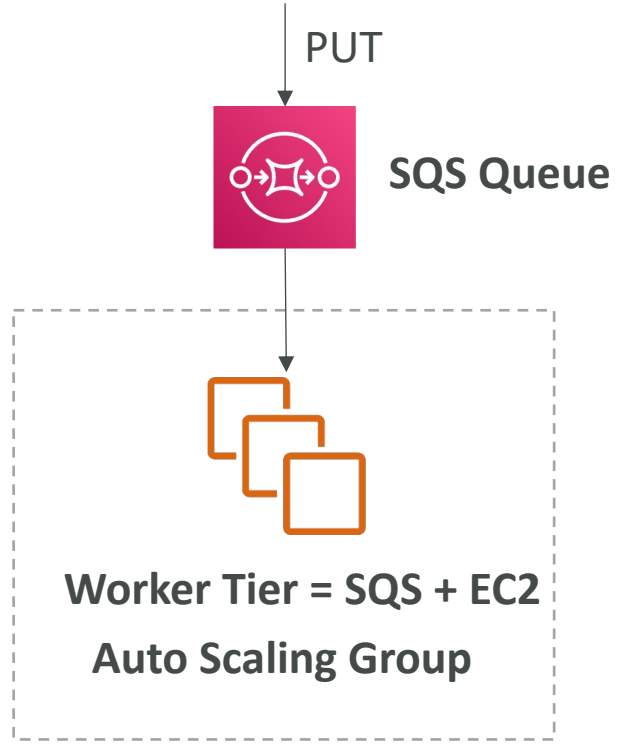
Single Instance
Great for dev



High Availability with Load Balancer
Great for prod



Worker Tier

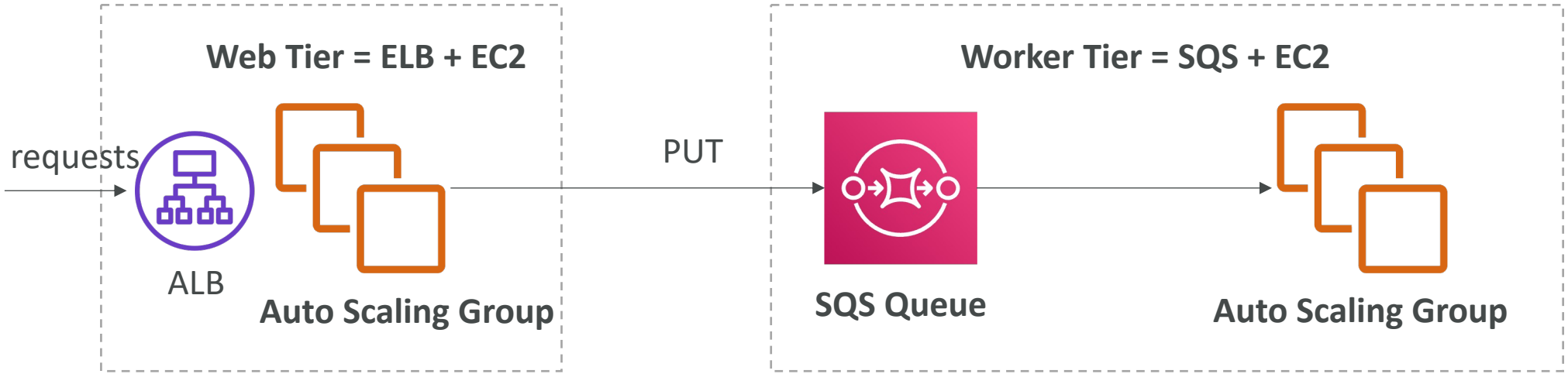


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Web Server vs Worker Environment

- If your application performs tasks that are long to complete, offload these tasks to a dedicated **worker environment**
- **Decoupling** your application into two tiers is common
- Example: processing a video, generating a zip file, etc
- You can define periodic tasks in a file **cron.yaml**

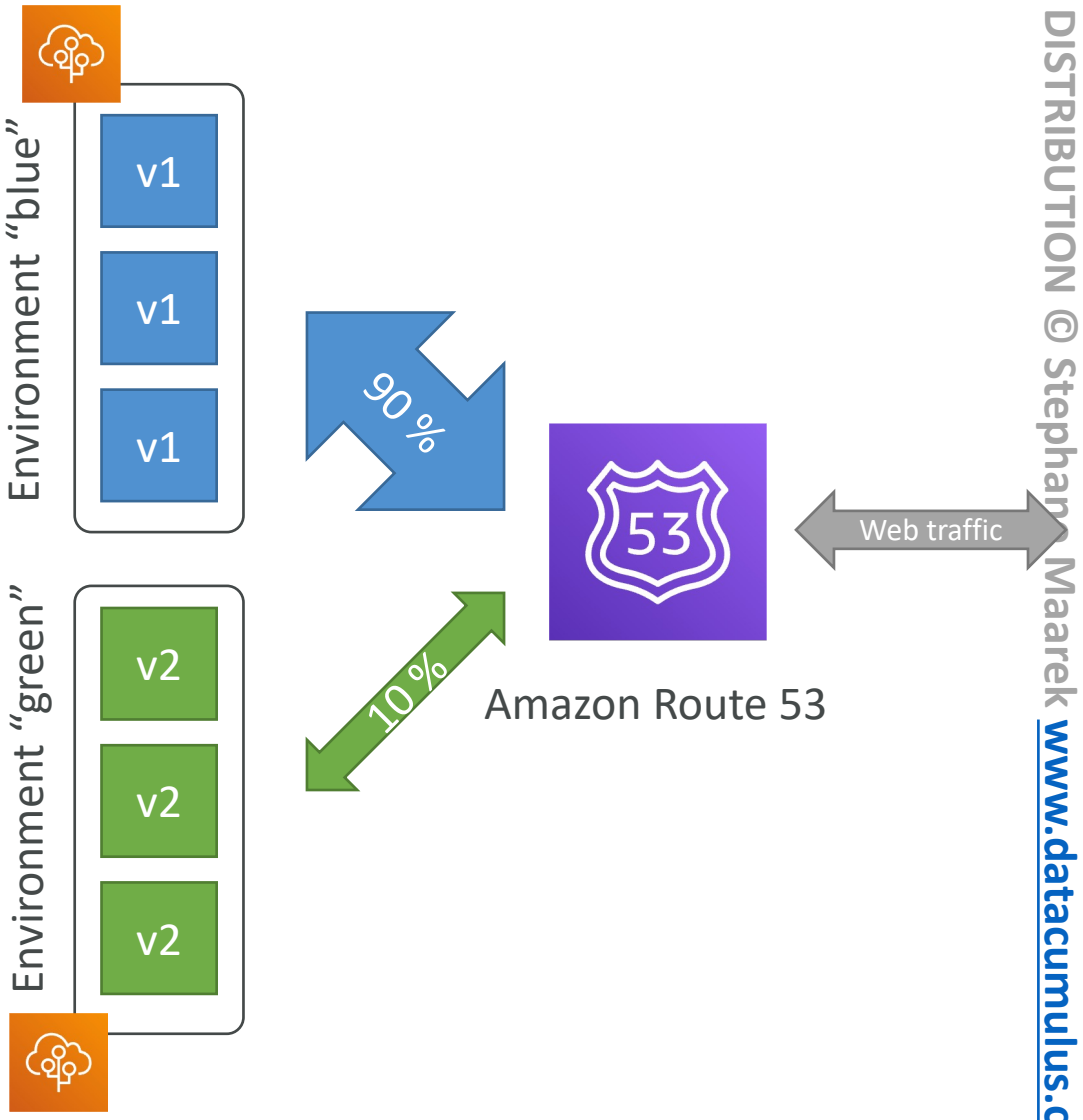


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Elastic Beanstalk Deployment Blue / Green

- Not a "direct feature" of Elastic Beanstalk
- Zero downtime and release facility
- Create a new "stage" environment and deploy v2 there
- The new environment (green) can be validated independently and roll back if issues
- Route 53 can be setup using weighted policies to redirect a little bit of traffic to the stage environment
- Using Beanstalk, "swap URLs" (DNS swap) when done with the environment test

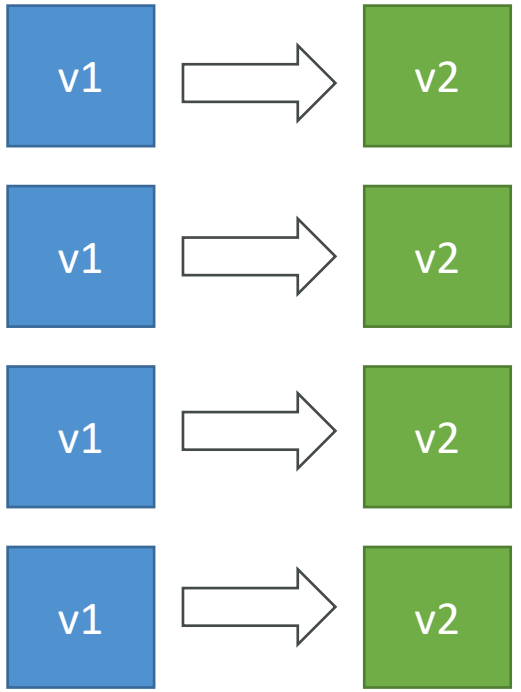


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AWS CodeDeploy

- We want to deploy our application automatically to many EC2 instances
- These instances are not managed by Elastic Beanstalk
- There are several ways to handle deployments using open source tools (Ansible, Terraform, Chef, Puppet, etc…)
- We can use the managed Service AWS CodeDeploy
- CodeDeploy can deploy to: **EC2, ASG, ECS & Lambda**

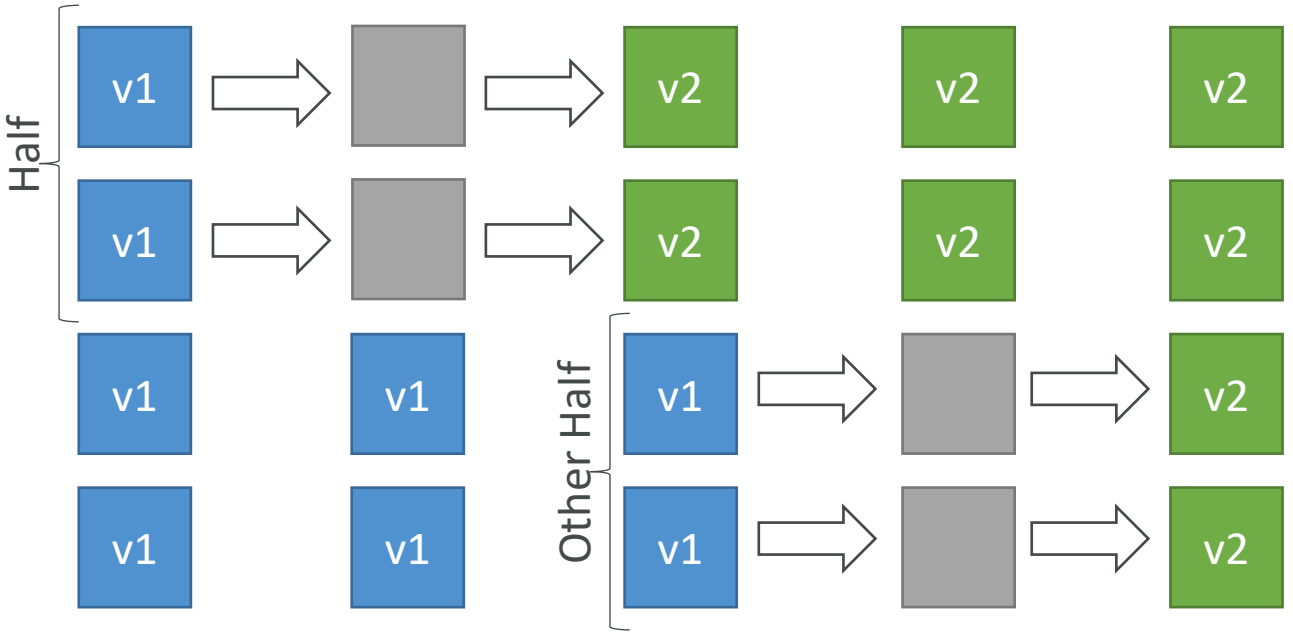


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CodeDeploy to EC2

- Define how to deploy the application using appspec.yml + deployment strategy
- Will do in-place update to your fleet of EC2 instances
- Can use hooks to verify the deployment after each deployment phase

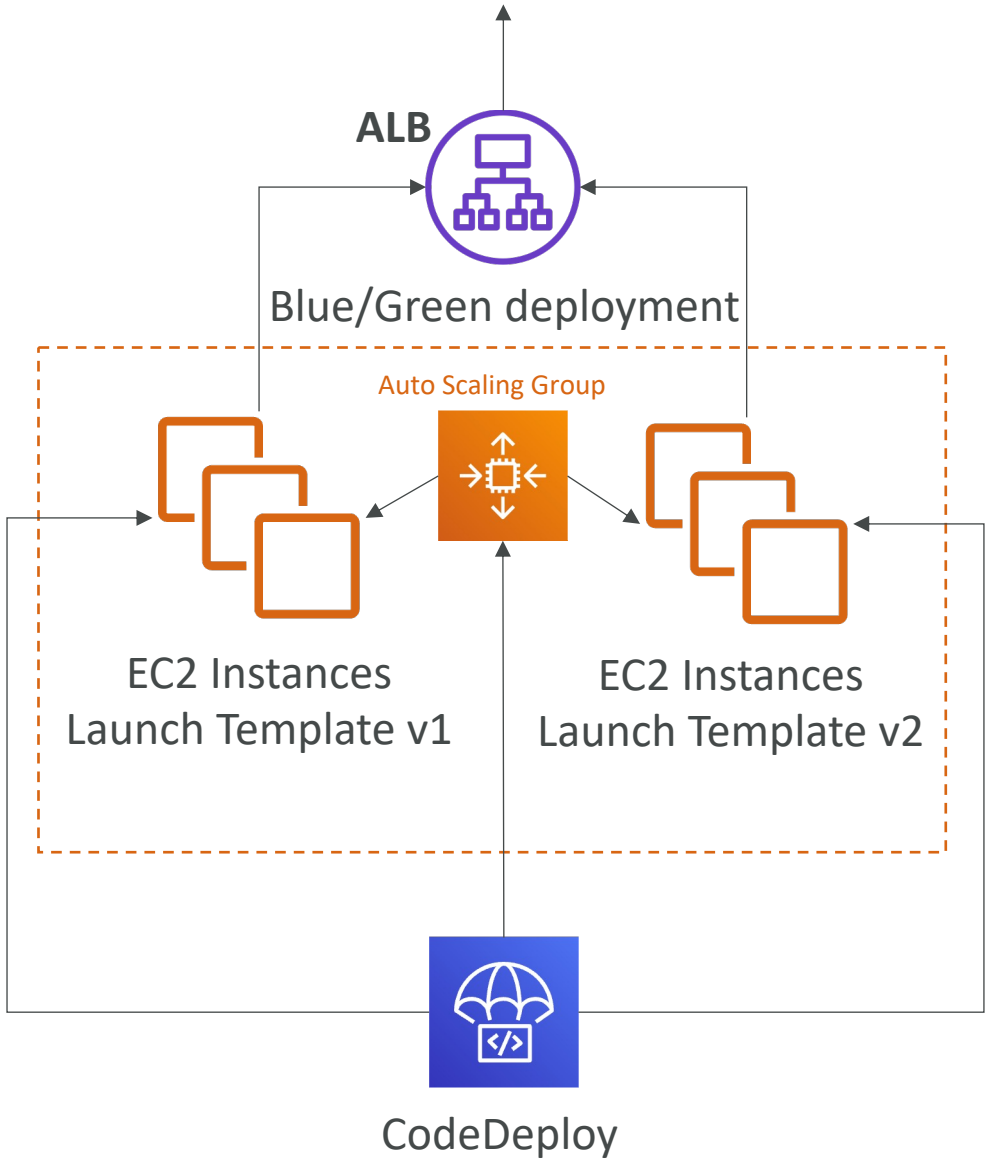


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CodeDeploy to ASG

- In place updates:
  - Updates current existing EC2 instances
  - Instances newly created by an ASG will also get automated deployments
- Blue / green deployment:
  - A new auto-scaling group is created (settings are copied)
  - Choose how long to keep the old instances
  - Must be using an ELB

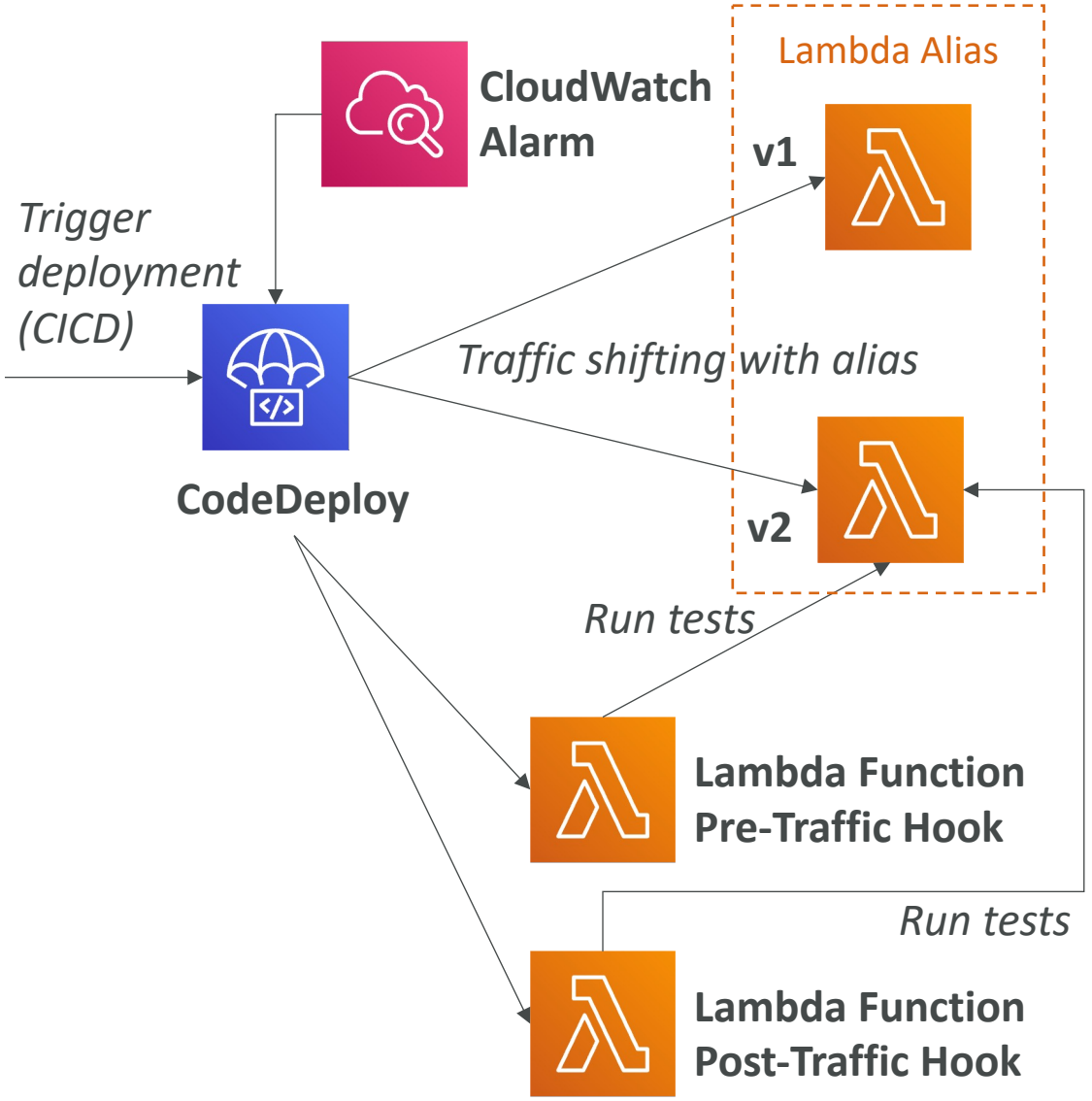


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CodeDeploy to AWS Lambda

- **Traffic Shifting** feature
- **Pre and Post traffic hooks** features to validate deployment (before the traffic shift starts and after it ends)
- Easy & automated rollback using CloudWatch Alarms
- SAM framework natively uses CodeDeploy

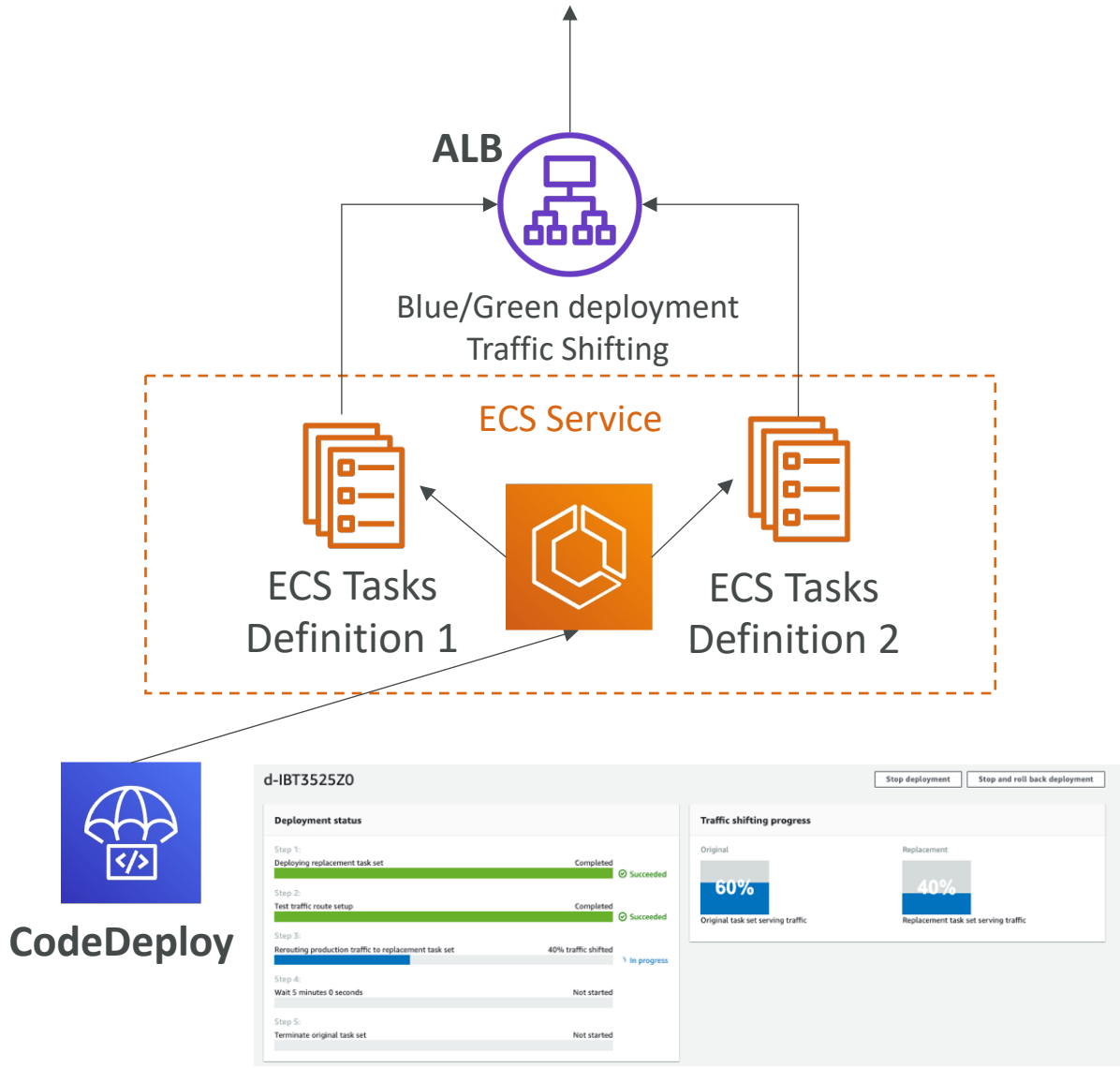


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CodeDeploy to ECS

- Support for Blue/Green deployments for Amazon ECS and AWS Fargate
- Setup is done within the ECS service definition
- A new task set is created, and traffic is re-routed to the new task test.
- Then if everything is stable for X minutes, the old task set is terminated (so you have time to notice issues)
- Supports Canary deployments (Canary10Percent5Minutes)



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AWS CloudFormation



- Infrastructure as code (IaC) in AWS
- Portability of stacks across multiple accounts and regions

- Backbone of the Elastic Beanstalk service
- Backbone of the Service Catalog service
- Backbone of the SAM (Serverless Application Model) framework

- Must-know service as a developer / sysops / devops

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Retaining Data on Deletes

- You can put a DeletionPolicy on any resource to control what happens when the CloudFormation template is deleted
- **DeletionPolicy=Retain**:
  - Specify on resources to preserve / backup in case of CloudFormation deletes
  - To keep a resource, specify **Retain** (works for any resource / nested stack)
- **DeletionPolicy=Snapshot:**
  - EBS Volume, ElastiCache Cluster, ElastiCache ReplicationGroup
  - RDS DBInstance, RDS DBCluster, Redshift Cluster
- **DeletePolicy=Delete (default behavior)**:
  - Note: for **AWS::RDS::DBCluster** resources, the default policy is Snapshot
  - Note: to delete an S3 bucket, you need to first empty the bucket of its content

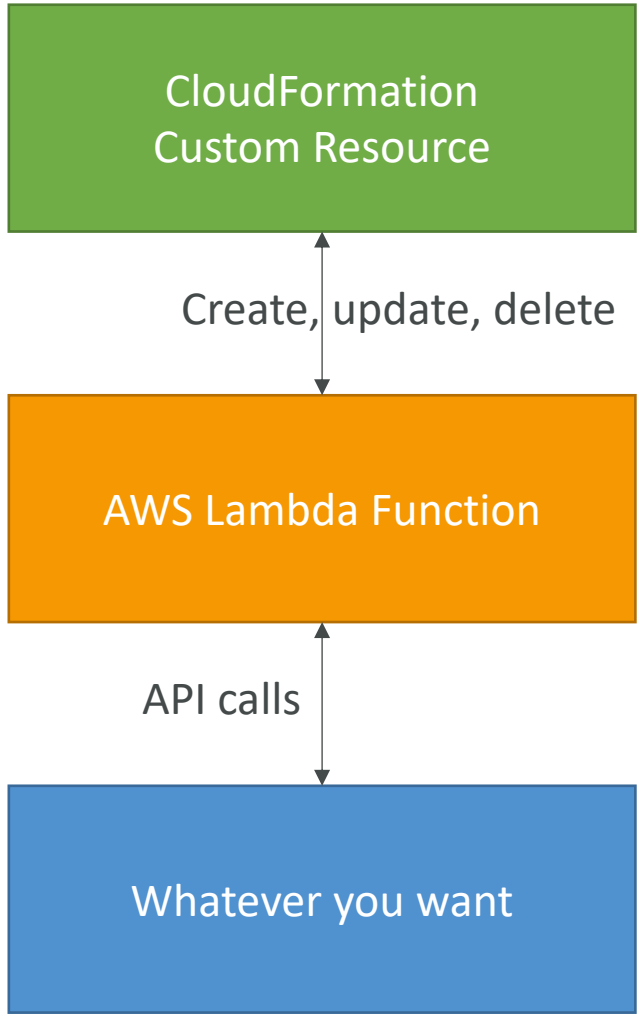
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CloudFormation Custom Resources (Lambda)

- You can define a Custom Resource in CloudFormation to address any of these use cases:

- An AWS resource is **not yet supported** (new service for example)
- An **on-premises resource**
- Emptying an S3 bucket before being deleted
- Fetch an AMI id
- Anything you want…!

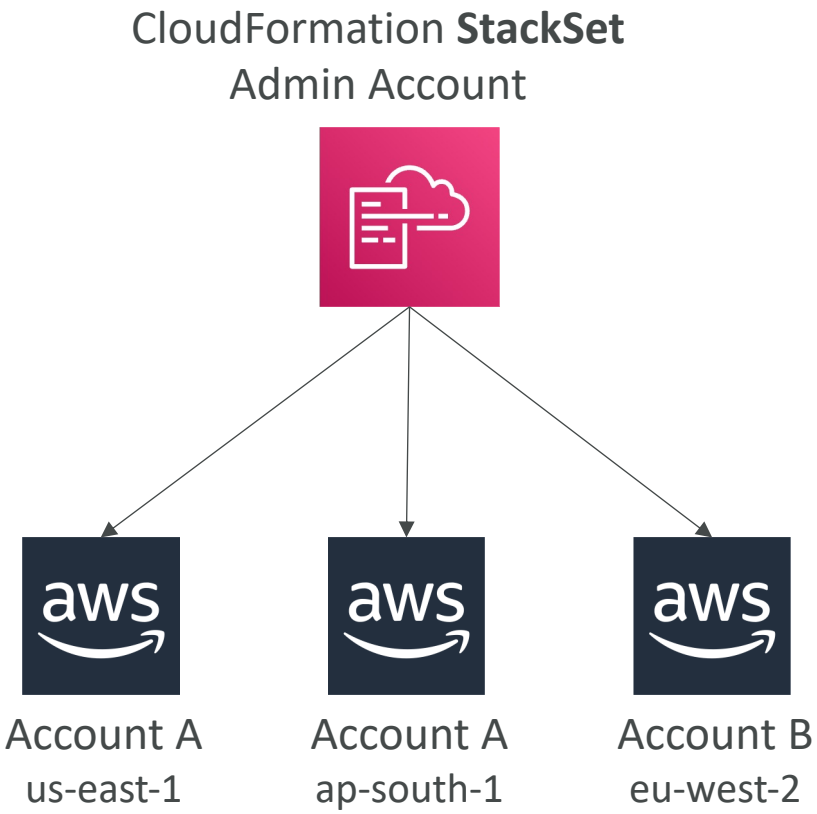


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CloudFormation – StackSets

- Create, update, or delete stacks across **multiple accounts and regions** with a single operation
- Administrator account to create StackSets
- Trusted accounts to create, update, delete stack instances from StackSets
- When you update a stack set, *all* associated stack instances are updated throughout all accounts and regions
- Enable **Automatic Deployment** feature to automatically deploy to accounts in AWS Organization or OUs



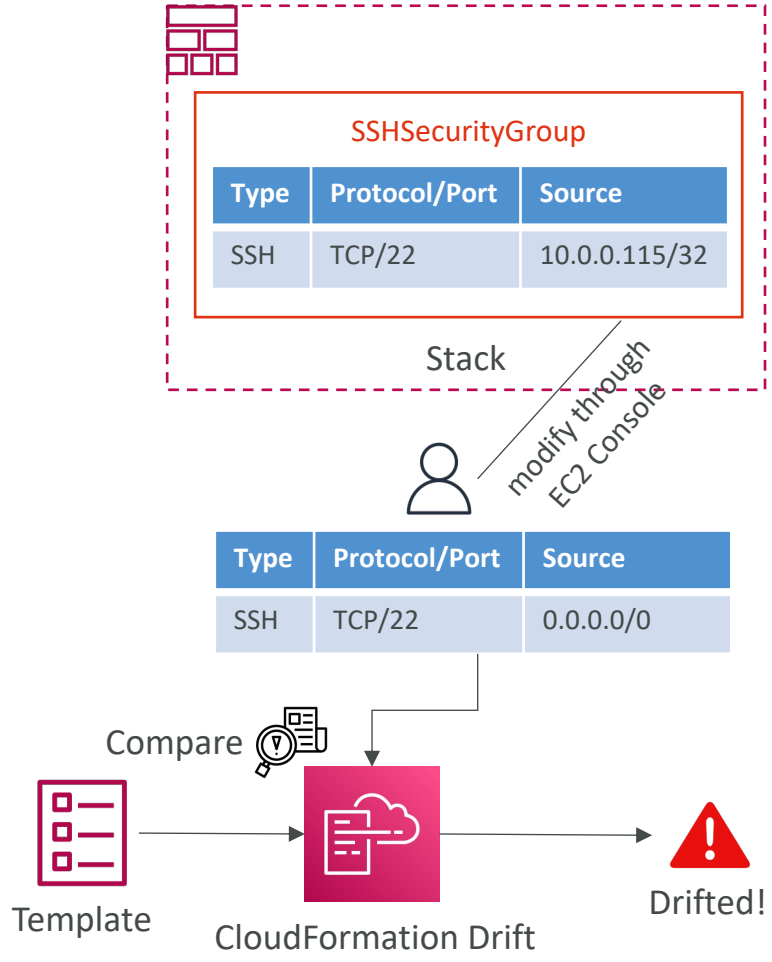
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CloudFormation Drift Overview

- CloudFormation allows you to create infrastructure
- But it doesn't protect you against manual configuration changes
- How do we know if our resources have **drifted**?

- We can use CloudFormation Drift!
- Detect drift on an entire stack or on individual resources within a stack



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CloudFormation – Integration with Secrets Manager

```yaml
Resources:
  # Secret resource with a randomly generated password in its SecureString JSON
  MyRDSDBInstanceRotationSecret:
    Type: AWS::SecretsManager::Secret
    Properties:
      GenerateSecretString:
        SecretStringTemplate: '{"username": "admin"}'
        GenerateStringKey: password
        PasswordLength: 16
        ExcludeCharacters: "\"@/\\"

  # RDS Instance resource. Its master username and password use dynamic references
  # to resolve values from Secrets Manager
  MyRDSDBInstance:
    Type: AWS::RDS::DBInstance
    Properties:
      DBInstanceClass: db.t2.micro
      Engine: mysql
      MasterUsername: !Sub "{{resolve:secretsmanager:${MyRDSDBInstanceRotationSecret}::username}}"
      MasterUserPassword: !Sub "{{resolve:secretsmanager:${MyRDSDBInstanceRotationSecret}::password}}"

  # SecretTargetAttachment resource which updates the referenced Secret with properties
  # about the referenced RDS instance
  SecretRDSDBInstanceAttachment:
    Type: AWS::SecretsManager::SecretTargetAttachment
    Properties:
      TargetType: AWS::RDS::DBInstance
      SecretId: !Ref MyRDSDBInstanceRotationSecret
      TargetId: !Ref MyRDSDBInstance
```

secret is generated

reference secret in RDS DB instance

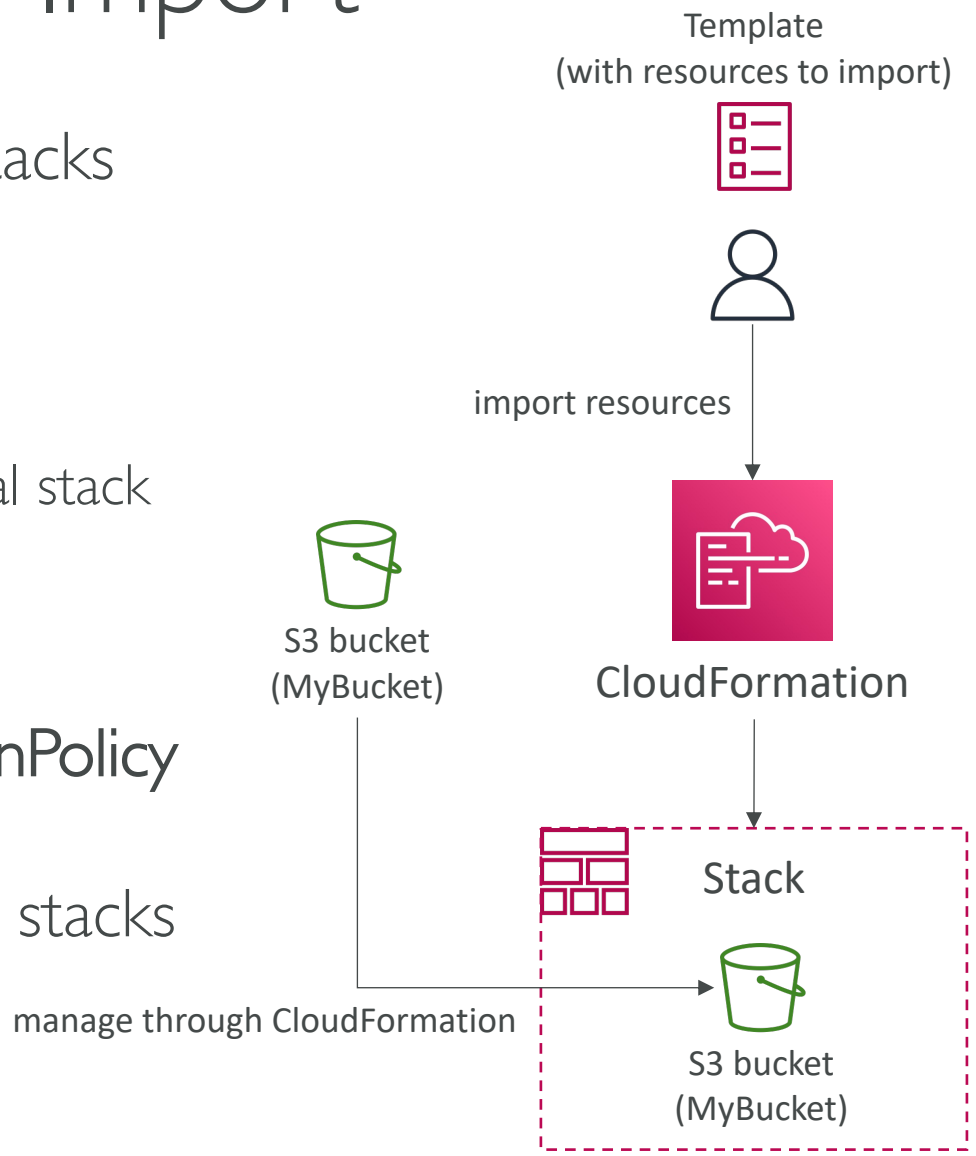
link the secret to RDS DB instance

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CloudFormation - Resource Import

- Import existing resources into existing/new stacks
- You don't need to delete and re-create the resources as part of a CloudFormation stack
- During import operation, you'll need
  - A template that describes the entire stack (original stack resources & target resources to import)
  - A Unique identifier for each target resource (ex. BucketName for S3 buckets)
- Each resource to import must have a **DeletionPolicy** attribute (any value) & Identifier
- Can't import the same resource into multiple stacks



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AWS Service Catalog



- Users that are new to AWS have too many options, and may create stacks that are not compliant / in line with the rest of the organization
- Some users just want a quick **self-service portal** to launch a set of **authorized products** pre-defined **by admins**
- Includes: virtual machines, databases, storage options, etc…
- Enter AWS Service Catalog!

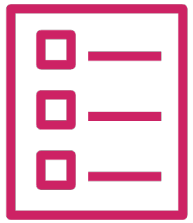
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Service Catalog diagram

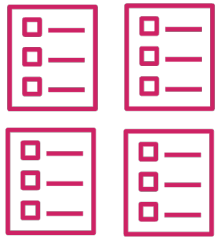
ADMIN TASKS

Product



CloudFormation Templates

Portfolio



Collection of Products

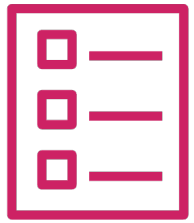
Control



IAM Permissions to Access Portfolios

USER TASKS

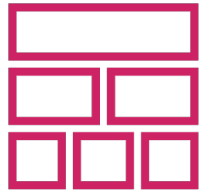
Product List



Authorized by IAM

launch

Provisioned Products



Ready to use
Properly Configured
Properly Tagged

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AWS Service Catalog



- Create and manage catalogs of IT services that are approved on AWS
- The "products" are CloudFormation templates
- Ex: Virtual machine images, Servers, Software, Databases, Regions, IP address ranges
- **CloudFormation helps ensure consistency, and standardization by Admins**
- They are assigned to Portfolios (teams)
- Teams are presented a self-service portal where they can launch the products
- All the deployed products are centrally managed deployed services
- **Helps with governance, compliance, and consistency**
- Can give user access to launching products without requiring deep AWS knowledge
- Integrations with "self-service portals" such as ServiceNow

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AWS SAM - Serverless Application Model

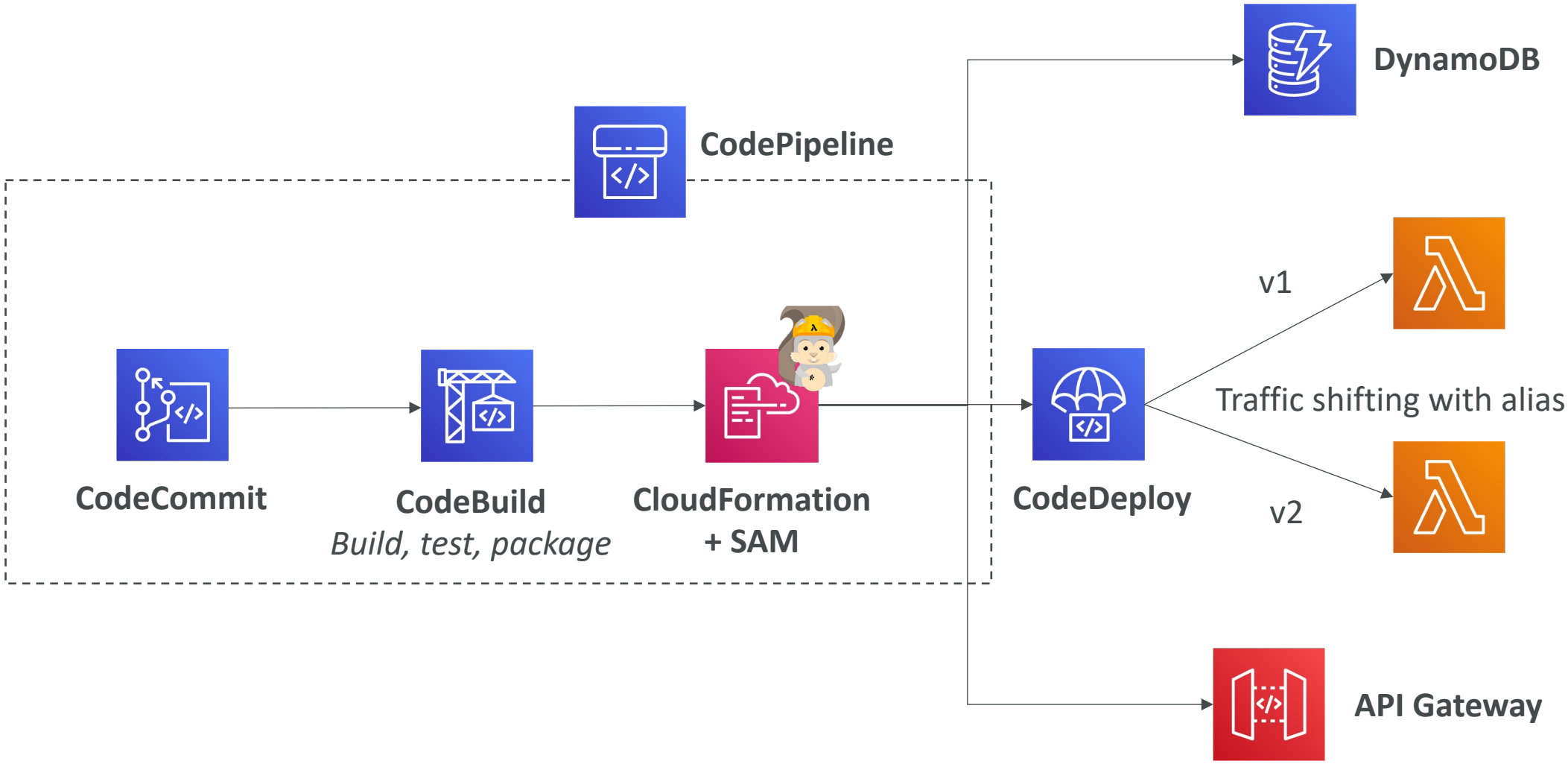


- SAM = Serverless Application Model
- Framework for developing and deploying serverless applications
- All the configuration is YAML code. Examples:
  - Lambda Functions (AWS::Serverless::Function)
  - DynamoDB tables (AWS::Serverless::SimpleTable)
  - API Gateway (AWS::Serverless::API)
  - StepFunction - State Machine (AWS::Serverless::StateMachine)
- SAM can help you to run Lambda, API Gateway, DynamoDB locally
- SAM can use CodeDeploy to deploy Lambda functions (traffic shifting)
- Leverages CloudFormation in the backend

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CICD Architecture for SAM



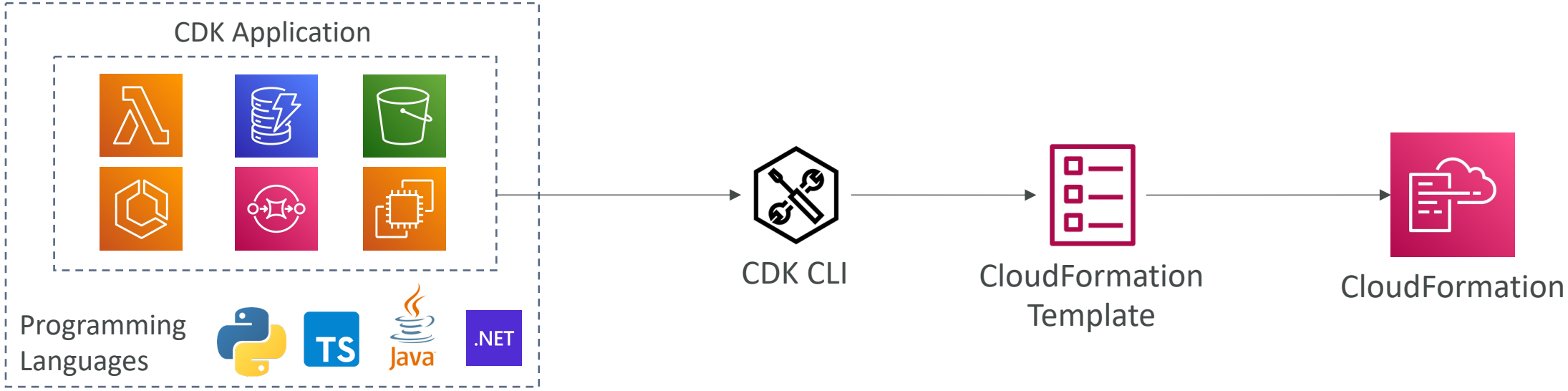
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AWS Cloud Development Kit (CDK)



- Define your cloud infrastructure using a familiar language:
  - JavaScript/TypeScript, Python, Java, and .NET
- The code is "compiled" into a CloudFormation template (JSON/YAML)
- You can therefore deploy infrastructure and application runtime code together
  - Great for Lambda functions
  - Great for Docker containers in ECS / EKS



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CDK Example

```
export class MyEcsConstructStack extends core.Stack {
  constructor(scope: core.App, id: string, props?: core.StackProps)
    super(scope, id, props);

    const vpc = new ec2.Vpc(this, "MyVpc", {
      maxAzs: 3 // Default is all AZs in region
    });

    const cluster = new ecs.Cluster(this, "MyCluster", {
      vpc: vpc
    });

    // Create a load-balanced Fargate service and make it public
    new ecs_patterns.ApplicationLoadBalancedFargateService(this, "My
      cluster: cluster, // Required
      cpu: 512, // Default is 256
      desiredCount: 6, // Default is 1
      taskImageOptions: { image: ecs.ContainerImage.fromRegistry("am
      memoryLimitMiB: 2048, // Default is 512
      publicLoadBalancer: true // Default is false
    });
  }
}
```

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AWS Systems Manager Overview



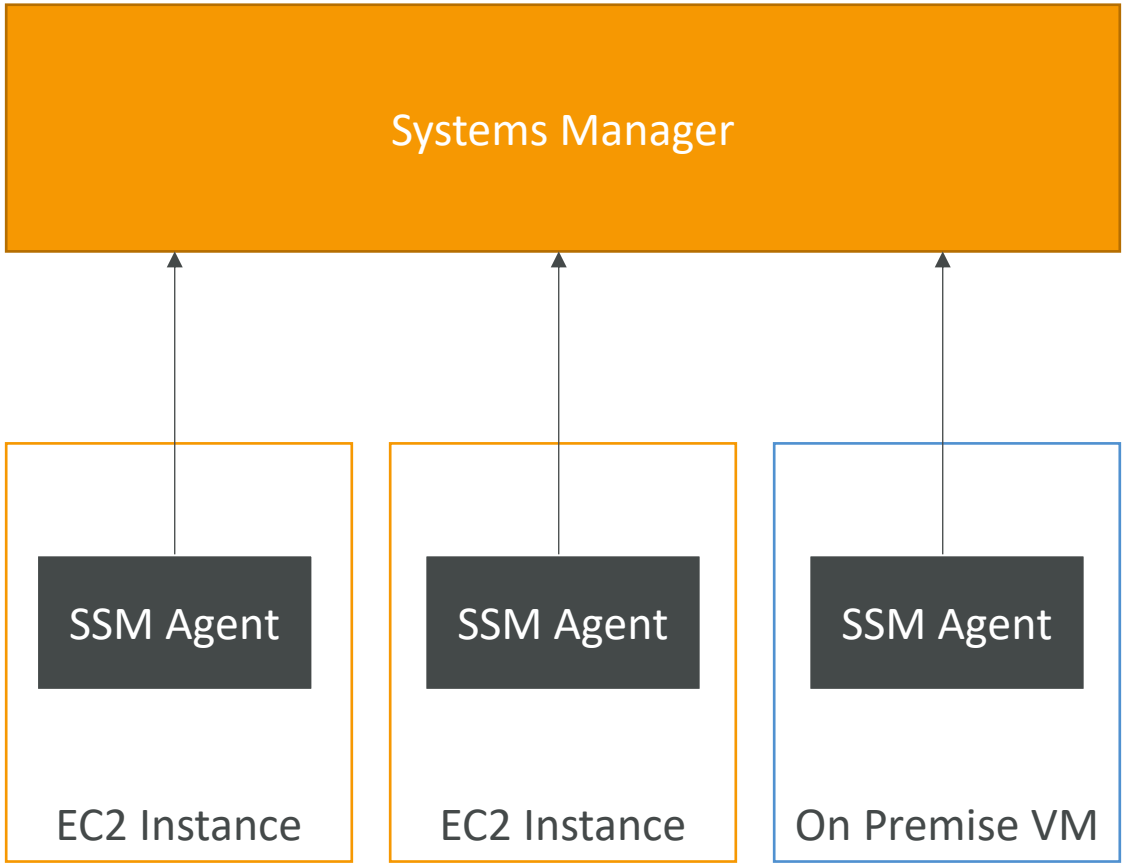
- Helps you manage your **EC2 and on-premises** systems at scale
- Get operational insights about the state of your infrastructure
- Easily detect problems
- **Patching automation for enhanced compliance**
- Works for both Windows and Linux OS
- Integrated with CloudWatch metrics / dashboards
- Integrated with AWS Config
- Free service

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How Systems Manager works

- We need to install the SSM agent onto the systems we control
- Installed by default on Amazon Linux AMI & some Ubuntu AMI
- If an instance can't be controlled with Systems Manager, it's probably an issue with the SSM agent!
- Make sure the EC2 instances have a proper IAM role to allow Systems Manager actions

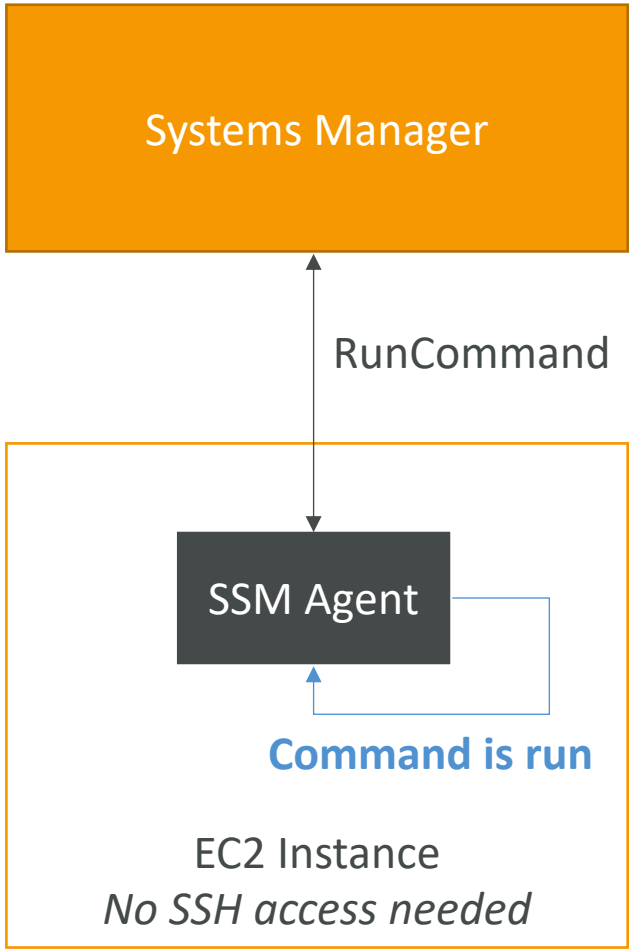


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AWS Systems Manager Run Command

- Execute a document (= script) or just run a command
- Run command across multiple instances (using resource groups)
- Rate Control / Error Control
- Integrated with IAM & CloudTrail
- No need for SSH
- Results in the console

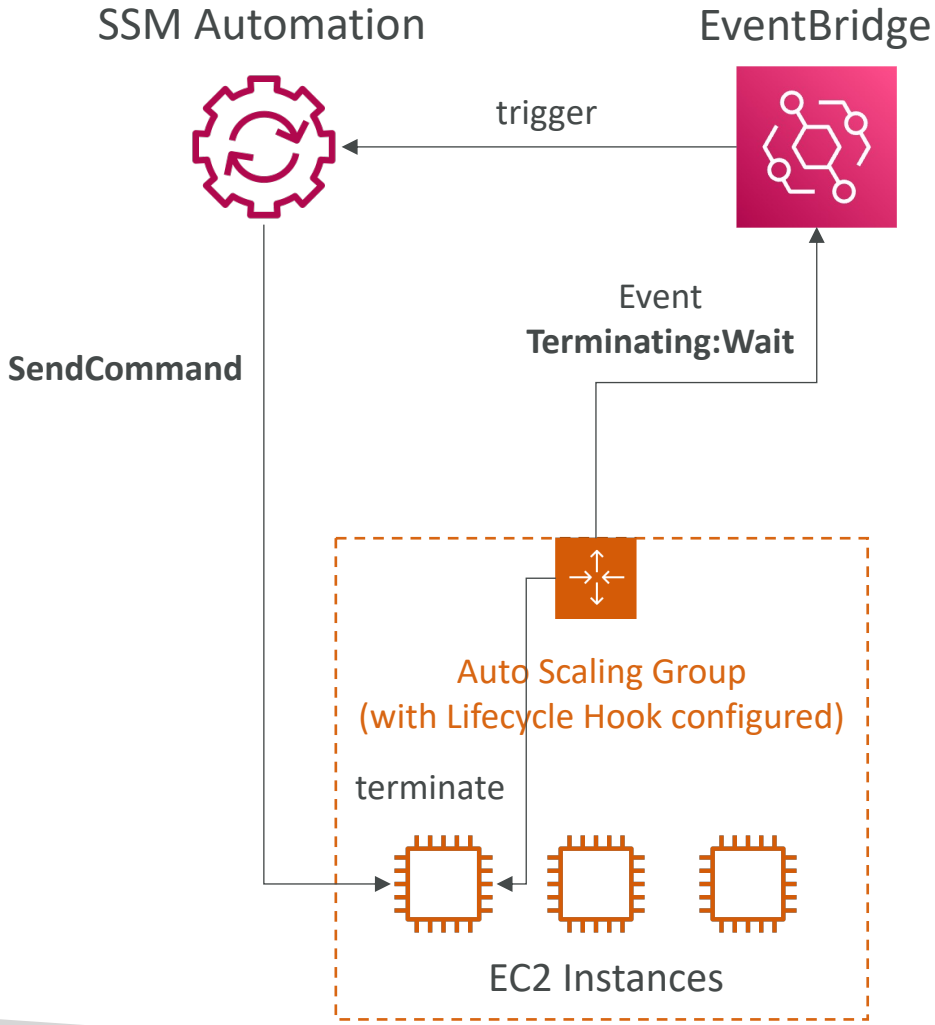


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AWS Systems Manager – Send Command before an ASG Instance is Terminated

- Perform any action before the ASG terminates an EC2 instance
- Create a ASG Lifecycle Hook that puts the instance in **Terminating:Wait** state
- Monitor the **Terminating:Wait** state using EventBridge
- Trigger a SSM Automation Document to perform the actions on the instance before termination

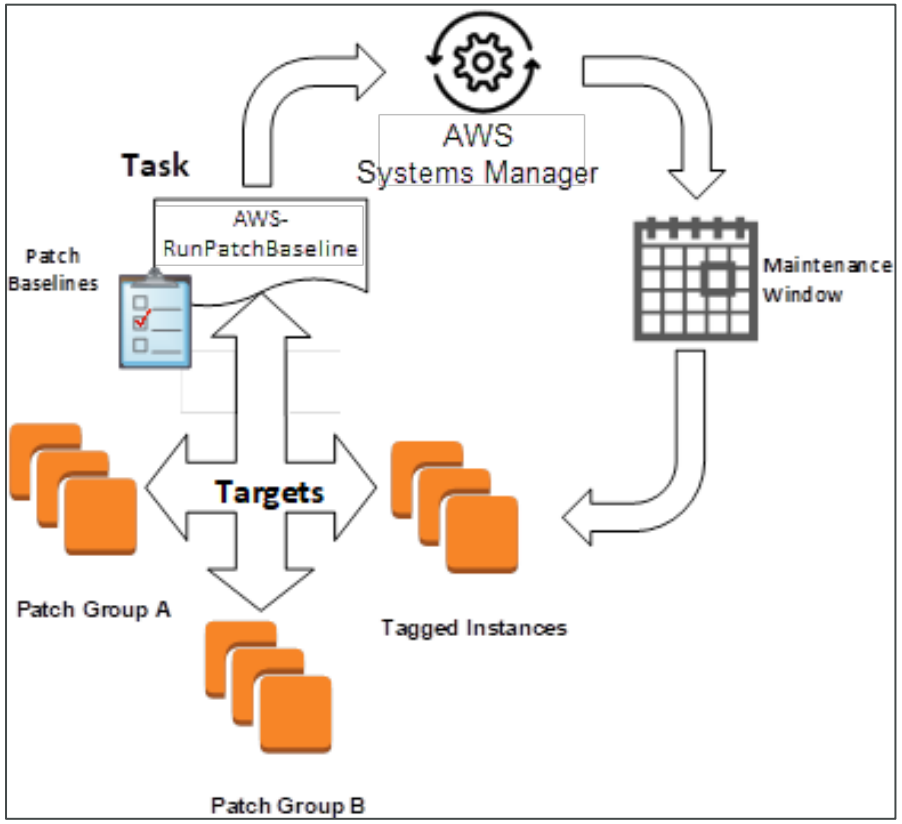


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Systems Manager Patch Managers – Steps

1. Define a **patch baseline** to use (or multiple if you have multiple environments)
2. Define patch groups: define based on tags, example different environments (dev, test, prod) – use tag **Patch Group**
3. Define **Maintenance Windows** (schedule, duration, registered targets/patch groups and registered tasks)
4. Add the **AWS-RunPatchBaseline Run Command** as part of the registered tasks of the Maintenance Window (works cross platform Linux & Windows)
5. Define **Rate Control** (concurrency & error threshold) for the task
6. Monitor **Patch Compliance** using **SSM Inventory**



https://aws.amazon.com/blogs/mt/patching-your-windows-ec2-instances-using-aws-systems-manager-patch-manager/

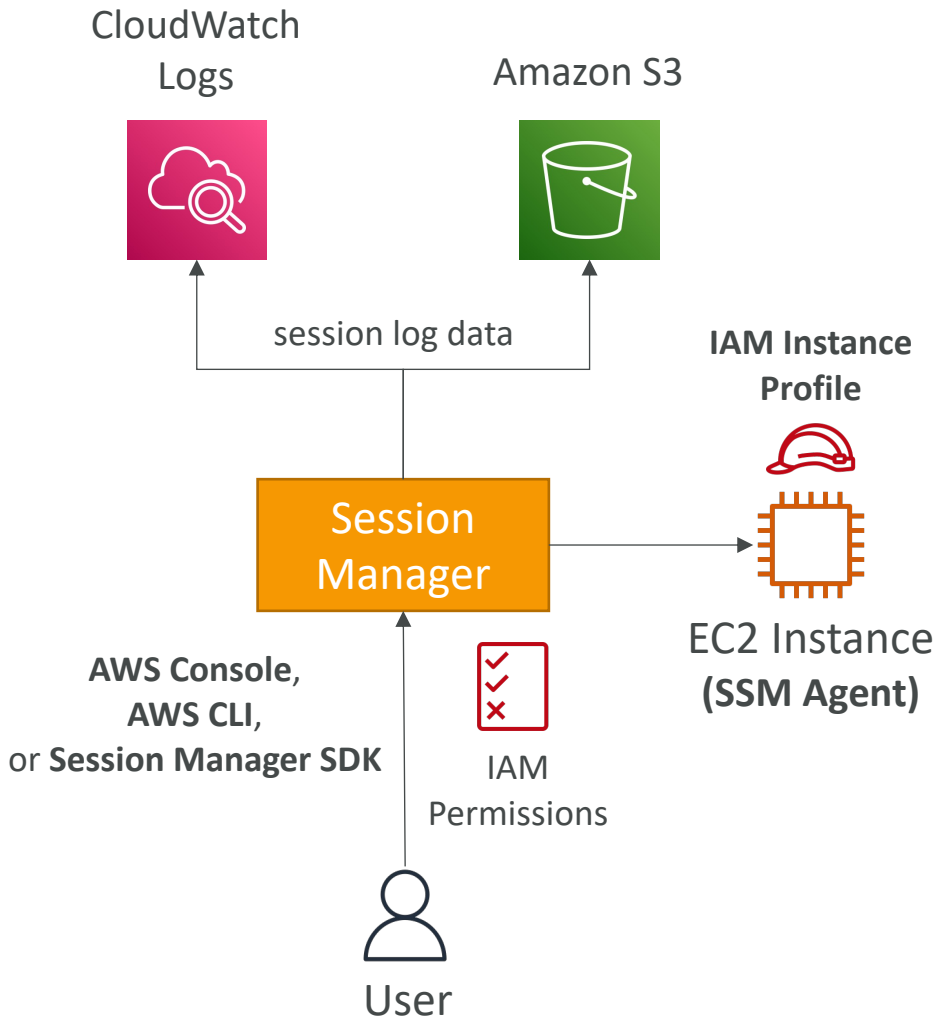
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Systems Manager Session Manager



- Allows you to start a secure shell on your EC2 and on-premises servers
- Access through AWS Console, AWS CLI, or Session Manager SDK
- **Does not need SSH access, bastion hosts, or SSH keys**
- Supports Linux, macOS, and Windows
- Log connections to your instances and executed commands
- Session log data can be sent to S3 or CloudWatch Logs
- CloudTrail can intercept **StartSession** events



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Systems Manager OpsCenter



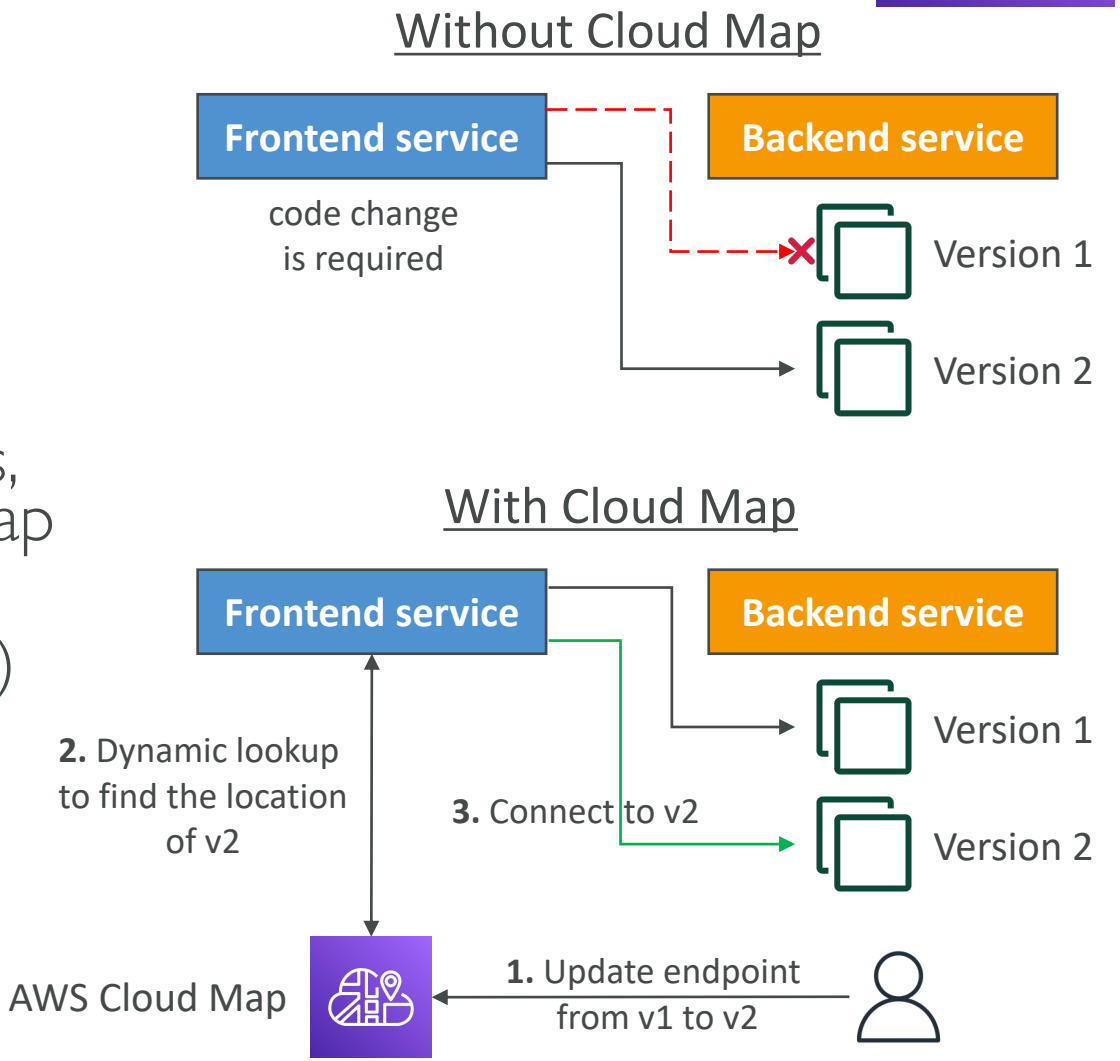
- Resolve Operational Issues (OpsItems) related to AWS resources
- **OpsItems** – issues, events and alerts
- Aggregates information to resolve issues on each OpsItems such as:
  - AWS Config changes and relationships
  - CloudTrail Logs
  - CloudWatch Alarms
  - CloudFormation Stack information
- Provides Automation Runbooks that you can use to resolve issues
- EventBridge or CloudWatch Alarms can create OpsItems

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AWS Cloud Map

- A fully managed resource discovery service
- Creates a map of the backend services/resources that your applications depend on
- You register your application components, their locations, attributes, and health status with AWS Cloud Map
- Integrated health checking (stop sending traffic to unhealthy endpoints)
- Your applications can query AWS Cloud Map using AWS SDK, API, or DNS



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